

PRODUCT AND CATEGORY MANAGEMENT

1.Database Design

1.1 Category Table

Attribute	Data Type	Key
Category_ID	NUMBER	Primary Key
Category_Name	VARCHAR2(100)	Unique Key
Description	VARCHAR2(255)	

1.2 Product Table

Attribute	Data Type	Key
Product_ID	NUMBER	Primary Key
Product_Name	VARCHAR2(100)	Not Null
Brand_ID	NUMBER	Foreign Key
Category_ID	NUMBER	Foreign Key
Price	NUMBER(10,2)	Not Null
Shade	VARCHAR2(50)	—
Skin_Type	VARCHAR2(50)	—
Expiry_Date	DATE	—
Stock	NUMBER	—

2. Primary and Foreign Key Relationship

Category → Product

Category.Category_ID → Primary Key

Product.Product_ID → Primary Key

Product.Category_ID → Foreign Key referencing Category.Category_ID

Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(100) UNIQUE,  
4     Description VARCHAR2(255)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Brand_ID NUMBER,  
5     Category_ID NUMBER,  
6     Price NUMBER(10,2) NOT NULL,  
7     Shade VARCHAR2(50),  
8     Skin_Type VARCHAR2(50),  
9     Expiry_Date DATE,  
10    Stock NUMBER,  
11    CONSTRAINT fk_product_category  
12         FOREIGN KEY (Category_ID)  
13         REFERENCES Category(Category_ID)  
14 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category VALUES (1, 'Face Care', 'Products for face care');
1 row created.

SQL>
SQL> INSERT INTO Category VALUES (2, 'Skin Care', 'Products for skin care');
1 row created.

SQL>
SQL> INSERT INTO Category VALUES (3, 'Hair Care', 'Products for hair care');
1 row created.

SQL>
SQL> INSERT INTO Category VALUES (4, 'Makeup', 'Makeup products');
1 row created.

SQL>
SQL> INSERT INTO Category VALUES (5, 'Body Care', 'Products for body care');
1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

4.2 Product Records

```
SQL> INSERT INTO Product
  2 VALUES (101, 'Face Wash', 201, 1, 250.00, 'White', 'Oily', DATE '2027-05-10', 50);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (102, 'Moisturizer', 202, 2, 350.00, 'Cream', 'Dry', DATE '2027-08-15', 40);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (103, 'Shampoo', 203, 3, 450.00, 'Clear', 'All', DATE '2028-01-20', 30);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (104, 'Lipstick', 204, 4, 299.00, 'Pink', 'All', DATE '2027-11-12', 25);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (105, 'Body Lotion', 205, 5, 399.00, 'White', 'Dry', DATE '2028-02-18', 35);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
1
```

```
Face Care
```

```
Products for face care
```

```
2
```

```
Skin Care
```

```
Products for skin care
```

```
CATEGORY_ID
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
3
```

```
Hair Care
```

```
Products for hair care
```

```
4
```

```
Makeup
```

```
CATEGORY_ID
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
Makeup products
```

```
5
```

```
Body Care
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
-----  
  BRAND_ID CATEGORY_ID    PRICE  
-----
```

```
SHADE
```

```
SKIN_TYPE
```

```
EXPIRY_DA
```

```
STOCK
```

```
-----  
          101  
Face Wash  
          201          1      250
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
-----  
  BRAND_ID CATEGORY_ID    PRICE  
-----
```

```
SHADE
```

```
SKIN_TYPE
```

```
EXPIRY_DA
```

```
STOCK
```

```
-----  
White  
Oily  
                                10-MAY-27          50
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
-----  
  BRAND_ID CATEGORY_ID    PRICE  
-----
```

```
SHADE
```

```
SKIN_TYPE
```

```
EXPIRY_DA
```

```
STOCK
```

```
-----  
          102
```

6. Product Update

```
SQL> UPDATE Product
  2 SET Product_Name = 'New Face Wash',
  3     Brand_ID = 210,
  4     Category_ID = 2,
  5     Price = 400.00,
  6     Shade = 'Natural',
  7     Skin_Type = 'Dry',
  8     Expiry_Date = DATE '2028-06-15',
  9     Stock = 60
10 WHERE Product_ID = 101;
```

1 row updated.

```
SQL> SELECT * FROM Product
  2 WHERE Product_ID = 101;
```

PRODUCT_ID

PRODUCT_NAME

BRAND_ID CATEGORY_ID PRICE

SHADE

SKIN_TYPE

EXPIRY_DA

STOCK

101

New Face Wash

210

2

400

PRODUCT_ID

PRODUCT_NAME

BRAND_ID CATEGORY_ID PRICE

SHADE

SKIN_TYPE

EXPIRY_DA

STOCK

7. Product Deletion

```
SQL> DELETE FROM Product
  2 WHERE Product_ID = 101;
```

1 row deleted.

8. Category Wise Product Reports

```
SQL> SELECT
  2     c.Category_Name,
  3     p.Product_ID,
  4     p.Product_Name,
  5     p.Price,
  6     p.Shade,
  7     p.Skin_Type,
  8     p.Stock
  9 FROM Category c
 10 JOIN Product p
 11 ON c.Category_ID = p.Category_ID
 12 ORDER BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

PRICE SHADE

SKIN_TYPE

STOCK

Body Care

105

Body Lotion

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

PRICE SHADE

SKIN_TYPE

STOCK

399 White

Dry

35

8.1 Product Count by Category

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Product_Count
  4 FROM Category c
  5 LEFT JOIN Product p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Body Care
1

Face Care
0

Hair Care
1

CATEGORY_NAME

PRODUCT_COUNT

Makeup
1

Skin Care
2